

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/270479485>

Discoursal flaws in Medical English abstracts: A genre analysis per research- and text-type

Article in *Text - Interdisciplinary Journal for the Study of Discourse* · January 1990

DOI: 10.1515/text.1.1990.10.4.365

CITATIONS

245

READS

2,965

1 author:



[Françoise Salager-Meyer](#)
University of the Andes

68 PUBLICATIONS 3,404 CITATIONS

SEE PROFILE

Discoursal flaws in Medical English abstracts: A genre analysis per research- and text-type*

FRANÇOISE SALAGER-MEYER

Never before has so much information been available to so many and never before have our lives depended so much on our ability to handle information successfully. We need to be able to search out what we require, to assess critically the ideas and facts offered to us and to make use of our findings.

(Marland, 1981: 9)

Abstract

Statement of the problem: *Medical English (ME) abstracts have lately been criticized for being uninformative, misleading and lacking internal structuring.*

Purpose: *In order to find out what their actual discourse structure is, we carried out a move analysis of 77 ME abstracts, written between 1986 and 1989 and drawn from 37 different medical journals.*

Methods: *The three fundamental medical text-types (research papers, case reports and review articles) and the four basic research-types (clinical, basic, epidemiological and operative) were considered. The selection and organization of moves, and the pattern of paragraph structuring were recorded in each abstract in order to assess the adequacy of its internal structure.*

Results: *Only 52% of the abstracts we studied are well structured: 62% from research papers, 67% from case reports and 17% from review articles. The major discoursal deficiencies of the remaining abstracts are 1) one fundamental move is missing; 2) two or more necessary moves are lacking; 3) illogical order in move presentation and 4) conceptual overlapping in paragraph structure.*

Conclusions: Because abstracts are the readers' first encounter with the text and because they assume a pivotal role in scientific development, medical researchers should pay a much closer attention to the way they structure their abstracts and medical journal editors to the abstracts of the papers they select for publication.

1. Introduction

In our increasingly busy world where researchers and academics are inundated by scholarly journals and books, efficient and effective literature searching is a necessity for the professional. Indeed, in the past twenty years, it has been established that physicians perceive periodicals to be the most valuable and available source for their on-going education and for disseminating new knowledge (Vollan, 1955; Harris, 1966; and Currie, 1975, 1976) but, because of the tremendous growth in the number of periodicals published and the interdisciplinary nature of research, no professional can take the time to survey the literature (Berry, 1981; Swales, 1981; Politser, 1987). The Louis Harris pool (1987) carried out for the New York Academy of Medicine and the Williamson et al. survey (1989) both reached the conclusion that the journal literature is seen as having an 'unmanageable' size. Too much time is needed to digest and synthesize what is relevant, valid and worth further attention (Huth, 1987 a, b). The formidable task of extracting the pertinent literature thus remains and grows (Mulrow, 1987; Hancock, 1989). The journal explosion is here to stay and what Berry wrote a few years ago (1981: 402) about the medical profession is likely to be true of any profession: 'if we do not retrace the steps of evolution, the most important entry requirement for future medical students will have to be proficiency in speed reading'. We could safely add: 'speed reading in English', the current scientific lingua franca and the world's major language of scholarship. In his editorial about English as an international language of medicine, Maher (1987) argues that fluency in reading skills is the minimum requirement not only for the comprehension of international journals but also for local in-house publications in countries where English is not the mother tongue. Thus, anything that helps the overburdened researcher to cope with the evermounting list of 'required' reading would definitely be welcome.

To keep up with what is being published in science and to follow in some way the hyperproduction of professional literature, scientists will have to rely more and more on abstracts as a short, concise, complete and accurate source of information (Ulijn, 1985; Kravac, 1987). Lock (1988) holds that most readers are content to read a paper's title and

abstract, casting an eye over the remaining sections. Indeed, because they represent an elaboration of the title and a crystallization of the whole article, abstracts (if well structured) give the readers an exact knowledge of the content of the article. As a pre-reading phase, they serve as a time-saving device that 1) introduces and arouses interest in the topic and, consequently, indicates to the readers whether the full text (or part of it) merits their further attention; 2) provides, when necessary, some language preparation for the text by mentioning key words and key concepts which will be dealt with in the article; 3) presents the article macropropositions and 4) help to understand the writers' purpose and the text structure. As a post-reading phase, abstracts help to consolidate or reflect upon what has been read and to relate the text to the learners' own knowledge or opinions. In short, abstracts have a pivotal role and should be the starting point, not an after-thought, of any professional reading.

Unfortunately, our personal experience has repeatedly shown that non-native readers of English, especially the less proficient ones, encounter great difficulties when they have to read scholarly paper abstracts. There are several explanations for this: 1) Because abstracts represent the highest level of conceptual sophistication at which writers and readers communicate, viz., peer group writing at specialist level. 2) Because of their communicative function, they strive for economy, avoid redundancy, show a high level of propositional density (a factor of difficulty which affects reading comprehension, cf. Kintsch and Keenan, 1973), and condense their informative content to the strict minimum (Graetz, 1985: 131). 3) Because of space constraints, abstracts lack *explicit* structuring. This makes abstracts even more difficult to understand because, as recent research on cognitive psychology and text and discourse analysis has shown, the reading process is altered and the end-product (comprehension) hindered when a text is poorly (or not) structured (Meyer, 1980, 1981; Marshall, 1984; Davies 1985). It can thus be claimed that abstracts make great demands on the readers' prior conceptual knowledge, on their linguistic competence (especially in the case of non-native speakers as well as on their command of the implicit rhetorical structures of this specific genre. In this respect, it is interesting to note that Huerta et al. (1986) remarked that Chilean students themselves mentioned abstract processing as one of their major problems in their scientific reading. This is very significant if we consider that the results of a survey the authors carried out showed 1) that the most important language skill for these students was technical reading comprehension (100% of preference), and 2) that almost 90% of the professors assign bibliographies in English.

Several studies on abstracts have been conducted in the past and have focused either on abstract reading (Hartley, 1988; Kieras and Bovair,

1981), on abstract writing (Cremmins, 1982; Krvavac, 1987), on the statistical analysis of their recurrent linguistic features (Gopnik, 1972, Guangfu, n.d.), or on their discoursal structures (Graetz, 1985; Harvey, 1986; Harvey and Horsella, 1988). It is especially interesting to note that 20 years ago, in an article published in the *Karger Gazette*, Ertl (1969) pointed out that abstracts all too often lack details that many readers need. He then suggested that abstracts be written in a tabular format which would allow for more detail, with clearer indications of the main structural components of papers. For aesthetic considerations and because such a format would take more space than the conventional 150 to 250-word abstract, this proposal has not been put to use by any journal (Huth, 1987a).

However, it would seem that Ertl's general principle has lately inspired editors of some leading medical journals. In fact, in the past two years a number of articles on medical English abstracts have appeared in the medical literature (Ad Hoc Working Group, 1987; Altman and Gardner, 1987; Huth, 1987a; Mulrow, 1987; Mulrow et al., 1988; Lock, 1988). They all show a growing concern, from the medical profession itself, for the way abstracts of clinical research papers and reviews are structured, as well as a plea for prompt corrective measures. Their main criticism is that most abstracts are uninformative, misleading and poorly structured. This is the reason why we decided to carry out a study on the discourse structure of Medical English abstracts which, as was said before, should be the 'step-number-one' bibliographical search in any professional reading.

2. Purpose, corpus and methodology

In order to discover what the actual internal structuring of Medical English abstracts is, we carried out a move analysis of 77 abstracts taken from clinical, basic, epidemiological and operative research¹ articles written between 1986 and 1989. The three major text-types have been considered, viz., research papers, case reports and reviews. Our corpus distribution (see Figure 1) reflects, to a certain extent, the importance given to each text-type in medical journals (Brewer, 1987).² A specialist informant, a Spanish-speaking medical researcher fluent in English, helped to classify the journals and the abstracts according to research- and text-type. Besides, in order to check for inter-observer variability, he independently carried out a move analysis of 25 abstracts randomly taken from our corpus. A cross-checking analysis of his results with ours revealed an inter-rater agreement in 86% of the cases. Finally, we could also discuss both the content and structure of some 'problematic' abstracts, for example, when the boundaries of one (or two) moves could not be readily determined.

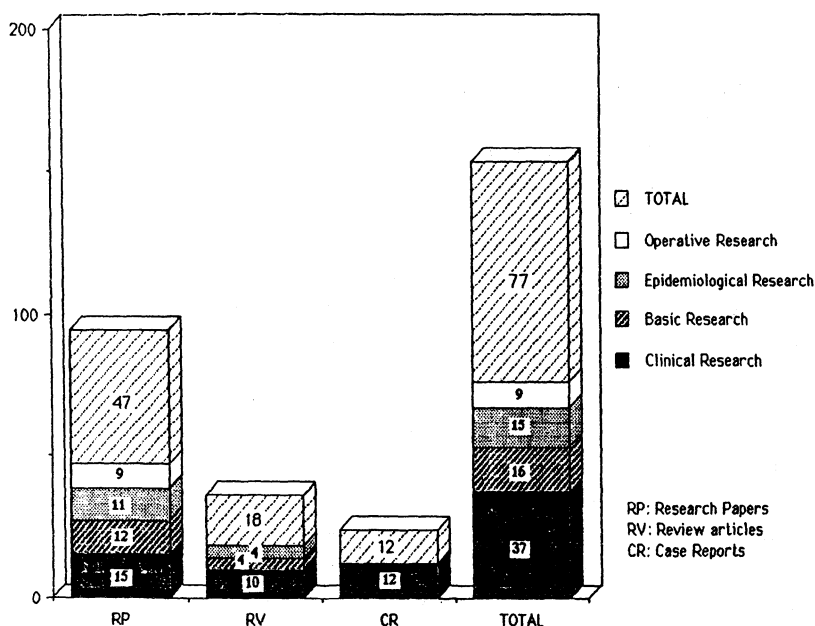


Figure 1. Corpus Distribution Per Text- and Research-Type

Of the 117 journals indexed in the *Abridged Index Medicus*, the journals with circulation greater than 50,000 were identified: 39 abstracts were taken from general readership journals (for example, *The Lancet*, *The New England Journal of Medicine*, *The American Journal of Medicine*, *Annals of Internal Medicine*) and 38 from 24 more specialized journals (for example, *Circulation*, *Kidney International*, *Journal of Chronic Diseases*, *Journal of Epidemiology and Community Health*, etc.). The size of our corpus is considered fairly adequate as a basis for making useful generalizations about the discourse organization in Medical English abstracts. However, reliability will be improved when the current findings are tested out on a further similar-sized (or larger) selection of Medical English abstracts.

Following on from the work of Swales (1981), McKinlay (1983) and Dudley-Evans (1986), we began this study by a general perusal of each abstract to get the feel of the overall organization and presentation. Then followed a preliminary identification, marking and ordering of the semantic units (or, as Swales calls them 'moves') that appeared. The number of paragraphs making up each abstract was also recorded.

To be considered *well structured*, an abstract should fulfill a certain number of discourse structure requirements which are closely related to

the basic intellectual structure (or macrostructures, cf. van Dijk, 1980) of the whole article:

1) The first one has to do with *move selection*: a well-structured abstract should state the four moves which are fundamental and obligatory in the process of scientific inquiry and patterns of thought, viz., purpose, methods, results (or data synthesis) and conclusions (optional in case reports).³ In fact, these four units are the ones which are generally required from contributors in most 'guidelines for authors' in scholarly journals, scientific style manuals, and 'rules for submitting abstracts' at international congresses. Kenny (1985) holds that these moves are a pattern of thought fundamental to many aspects of human communication and of significance to information structuring in general. However, the 'purpose' and 'methods' moves are sometimes merged in one sentence and thus form a single move. In such a case, a three-move pattern abstract will be considered well-structured.

2) The second requirement deals with the *semantic organization of moves*: these should be presented in a logical way or, as van Dijk and Kintsch (1983) express: 'in their logical order'. As Hall et al. (1986) recommend, 'the sequence of information should represent the progression of ideas and should be consciously organized' (for example, the 'results' move should not precede the 'purpose' move).⁴

3) The third requirement is concerned with the more formal level of *paragraph structuring*: if the abstract is made up of more than one paragraph the physical paragraphs should correspond to conceptual ones. Although more than one concept can perfectly be expressed in a single physical paragraph, no conceptual overlapping should be observed from one paragraph to another.

By contrast, *poorly organized abstracts* are those which are deficient at any (or all) of the above mentioned levels, i.e., those which are not presented as a coherently argued and unified research study and thus evidence the writers' lack of awareness of appropriate discourse structure. Thus, the internal structuring of our 77 abstracts will be 'tested' against the above mentioned qualifications of a well-structured abstract at the move selection, move organization and paragraph structuring levels.

3. Results

3.1. *Well structured abstracts*

Out of 77 abstracts in the three research-types, only 40 (52%) conformed to the three above mentioned criteria for well-structured abstracts: 29

research papers, 8 case reports and 3 reviews. Table 1 presents our results per research and text-type. Moreover, as Table 2 indicates, 19 abstracts encapsulate a three-move pattern, 18 a four-move structure, two a five-move patterning and one an eight-move organization. Because some differences (mainly due to text-types) could be observed in the overall patterning of these 40 abstracts, we shall now present our findings per text-type.

3.1.1. *Research paper abstracts.* All but three well-structured research paper abstracts follow a three- or four-move pattern, the basic core structure of three-moves being 'purpose', 'results' and 'conclusions'. Optional moves such as 'statement of the problem' and 'recommendations' have been observed in eight cases. It must be noted, however, that the content of certain moves varies somewhat according to research-type. In clinical, basic and operative research, the first move is mostly a three-stage move which briefly describes the population (or cohort) features, the methods used and the tests performed, whereas in epidemiological research, it is basically a four-stage move which states the objective, setting, duration and corpus of the study. Moreover, the last move of research paper abstracts from operative research is more a recommendation than a real conclusion.

3.1.2. *Review article abstracts.* The three well-structured review abstracts follow a four move pattern which states the objectives of the study (with the corpus as a sub-move), the data synthesis (that is, the results), the conclusion and the recommendations.

3.1.3. *Case report abstracts.* The eight well-structured clinical case reports conform to a three- or four-move pattern. The basic three-move core structure includes the case presentation, the findings and a comment about the particular case(s). When present, the fourth move is either a 'statement of the problem' or a 'suggestion'. It is to be mentioned that the 'case presentation' is generally a two-stage move which includes either the most frequent findings or the tests performed on the patient(s).

We shall now present our findings with respect to the *paragraph structuring* (the third element of a well-structured abstract) of the 40 well-structured abstracts. The great majority of them (34, that is, 85%) are made up of one paragraph only. The remaining six abstracts are constructed with three, four or eight paragraphs. The abstracts made up of three paragraphs (three clinical research papers from *The New England Journal of Medicine*) are undivided by headings, while the three abstracts structured with four or eight paragraphs are explicitly marked, that is,

Table 1. *Frequency of well- and poorly-structured abstracts per text- and research-type*

Research type	Text-type							
	RP N=47		RV N=18		CR N=12		Total N=77	
	WS	PS	WS	PS	WS	PS	WS	PS
Clinical research N=37	11 73%	4 27%	1 10%	9 90%	8 67%	4 33%	20 54%	17 46%
Basic research N=16	8 67%	4 33%	0 0%	4 100%	— —	— —	8 50%	8 50%
Epidemiological research N=15	4 36%	7 64%	2 50%	2 50%	— —	— —	6 40%	9 60%
Operative research N=9	6 67%	3 33%	—	—	—	—	6 67%	3 33%
Total	29 62%	18 38%	3 17%	15 83%	8 67%	4 33%	40 52%	37 48%

Abbreviations used: RP: Research Papers; RV: Review articles; CR: Case Reports; WS: Well-structured; PS: Poorly-Structured.

Table 2. Number of Moves in the 40 well-structured abstracts

N° of moves	RP WS=29	RV WS=3	CR WS=8	Total WS=40
3 moves	Clin. Res. 4 Bas. Res. 3 Epid. Res. 4 Oper. Res. 3 Total 14	— — — — —	Clin. Res. 5	19 (48%)
4 moves	Clin. Res. 5 Bas. Res. 5 Oper. Res. 2 Total 12	Clin. Res. 1 Epid. Res. 2 Total 3	Total 5 Clin. Res. 3	18 (45%)
5 moves	Clin. Res. 1 Oper. Res. 1 Total 2	— — —	Total 3 — —	2 (5%)
8 moves	Clin. Res. 1 Total 1	—	—	1 (2%)
Total	29 WS abstracts of 3, 4, 5 or 8 moves	3 abstracts of 4 moves	8 abstracts of 3 and 4 moves	40 abstracts of 3, 4, 5 or 8 moves

Abbreviations used: RP: Research Papers; RV: Review articles; CR: Case Reports; WS: Well-Structured; Clin. Res: Clinical Research; Bas. Res: Basic Research; Epid. Res: Epidemiological Research; Oper. Res: Operative Research.

their paragraphs are preceded by specific headings. When four paragraphs are involved (two clinical research papers from *The American Journal of Medicine*), the headings are: 'purpose', 'methods', 'results' and 'conclusion.' As for the eight-paragraph structure (one clinical research paper from *The British Journal of Medicine*), it is worthwhile mentioning that it is the only abstract which strictly adheres to the guidelines suggested by the Ad Hoc Working Group: its eight paragraphs are clearly marked by the recommended headings ('objective', 'design', 'setting', 'subjects', 'intervention', 'end point', 'measurements and main results' and 'conclusion') and it is written in a point rather than the narrative form.

3.2. *Poorly structured abstracts*

The remaining abstracts, that is, those which violate conventional order, suffer from different types of discoursal flaws which are identified and introduced below in decreasing order of frequency.

37 of the 77 abstracts we studied (48%) have been found to be poorly structured: 18 from research papers, 15 from reviews and 4 from case reports. This means that, out of the whole corpus, 83% of abstracts from reviews, 38% from research papers and 33% from case reports present inconsistencies at the move and/or paragraph levels (see Table 3). We shall now present, in decreasing order of frequency, the discoursal pitfalls observed at the macro-level a) in move patterning (selection and organization) and b) in paragraph structuring of these 37 poorly structured abstracts.

3.2.1. *In move patterning.* 1. The leading cause of discoursal deficiency at the move level (see Table 3) is the absence of a fundamental semantic unit, for example, 'purpose' or 'conclusion'. This has been observed in ten research papers and four reviews abstracts. The absence of the 'comment' move has been noted in two case report abstracts. This discoursal flaw is the reason why almost half of the poorly structured abstracts have been classified as such. 2. In 27% of the poorly structured abstracts, two or more obligatory moves are missing: 'purpose' and 'conclusion' in six review abstracts, or 'purpose' and 'results' in one research paper abstract. What is more, in three instances (one research paper and two review abstracts), one move only ('results' or 'data synthesis') could be identified. 3. Either an illogical order in move organization (for example, the 'results' precede the 'purpose'; the 'conclusion' appears as the first move), or an awkward 'move recycling', that is, the same move is split between other moves (for example, some results are presented, then the conclusion and then the remaining results) has been noted in five abstracts (two reviews, two case

Table 3. *Discoursal deficiencies in the 37 poorly-structured abstracts*

Discoursal flaws	Text-type		Total	
	RP N = 47	RV N = 18	CR N = 12	% over 77 abstracts
One move missing	10 missing: conclusions 7 purpose 3	4 missing: conclusion 1 purpose 3	2 missing: comments 2	16 43% 21%
Two (or more) moves missing	2 missing: purpose 1 + results 1 purpose + methods + conclusion 1 1: conclusion precedes results 5	8 missing: purpose + concl. 6 purpose + methods + conclusion 2 2: results precede purpose 1	0	10 27% 13%
Illogical order			2: 2 move 'recycling'	5 14% 6%
Conceptual overlapping			0	6 16% 8%
Total	18 RP (38%)	15 RV (83%)	4 CR (33%)	77 100% 48%

Abbreviations used: RP: Research Papers; RV: Review articles; CR: Case Reports.

reports and one research paper). Such an erratic presentation accounts for 14% of the discursal flaws observed in poorly structured abstracts.

3.2.2. *In paragraphing structuring.* 31 of the 37 poorly structured abstracts are made up of one paragraph. In the remaining six abstracts (16%) 'organized' with two or three paragraphs, a conceptual overlapping could be observed. For example — and this is a particularly frequent situation in *The New England Journal of Medicine* — the move 'results' or 'patients and methods' overlaps over the first and second paragraph.

4. Discussion and suggestions

Hall et al. (1986) hold that engineering English abstracts neatly 'encapsulate the whole research four-move pattern' (unfortunately, the authors do not indicate either the size of the nature of the abstracts they studied). Our results with Medical English abstracts are not consistent with such a finding. These authors go on to argue that students arrive at this four-move pattern themselves by examining texts. If it is true that we learn to write by writing and by examining (or simply reading) *good models*, medical researchers should then pay much closer attention to the way they write their abstracts. Indeed, the fact that 48% of the abstracts we analyzed have been classified as poorly-structured suggests that the discourse organization of Medical English abstracts should promptly be improved, especially if we bear in mind the deleterious effect a poorly structured text has on reading comprehension.

Whilst emphasizing that the following statements are entirely speculative and need to be corroborated by further research, it would however seem reasonable to assume that the rhetorico-structural deficiencies mentioned in section 3.2 render a text more difficult to read. Not only could they make reading a time-consuming, tedious, fragmented, and unsatisfactory enterprise, but also prevent the smoothness of the reading process and lead to the reader's erratic performance. In addition, as Mulrow et al. (1987) so rightly set forth when referring to review abstracts, those in which either the purpose or the conclusion is missing do not allow the reader to *efficiently* assess the breadth and comprehensiveness of the whole article. We could speculate that this holds true as well for case report and research paper abstracts: discursively deficient research paper or case report abstracts do not permit the reader to *quickly* ascertain the potential relevance of a case report or the scientific soundness and generalizability of a clinical trial. Indeed, several studies have shown that rhetorical organization of a text affects comprehension (Mandler, 1978; Meyer,

1977a, b; Meyer et al., 1980; Carrell, 1985). Carrell (1987: 35) even recommends considering the rhetorical structure of a text and the transparency of the signalling of that rhetorical structure — both of which have been shown to affect reading comprehension —, as a key factor in any readability formula. Each one of the above mentioned discoursal flaws in isolation may not in itself make an abstract too difficult to read. It is when an abstract contains several of these factors that serious problems of readability will very likely arise. However, all these observations, we repeat, remain to be tested by careful psycholinguistic evidence.

Earlier in this paper, reference was made to the differences in move patterns observed in the three text-types and, to a lesser extent, in the four research-types. These differences are mainly attributable to the communicative function and inherent nature of each text- and research-type. For example, case reports, because they are observations made on one or a few patients only, can seldom state conclusions which would be generalizable to the universe of such patients (a 'comment' or a 'discussion' is given instead); the setting and duration of a study are important data in epidemiological research, whereas they may be optional in clinical research; in review articles (which generally state the results of bibliographical studies), data identification, selection, and validity assessment are fundamental moves. Our results also show that case report and research paper abstracts are relatively well-structured as compared with review abstracts: indeed, more than half of them reflect the whole article macropropositions. This is very likely due to the fact that for over 30 years journal editors have insisted that research papers be clearly formatted for the convenience of readers: the linguistically-marked IMRAD formal structure, although it had been termed in the past a straightjacket around the author (Lock, 1988), is now adopted by over 300 medical journals (International Committee of Medical Journal Editors, 1988). Case reports, too, present clearly-labeled macrostructures ('introduction', 'case report' and 'discussion/comment') although their format has not been stipulated as clearly as it has been for research papers. By contrast, review abstracts are the texts which least conform to the standardized guidelines based on principles adopted from literature concerning scientific review processes and information synthesis. In fact, none of them specified the suggested six-part tabular format: 1. purpose, 2. data identification, 3. study selection, 4. data extraction, 5. results of data synthesis and 6. conclusions. What is more, our results show that only six review abstracts stated the conclusion or the purpose of the study. The results obtained from reviews clearly corroborate other researchers' findings. It has been shown that medical review articles are scientifically unsound, often suggestive and inefficient (Light and Pillemar, 1984), that strategies

for identifying and selecting information are rarely defined and that collected information is reviewed haphazardly with little attention to systematic assessment of quality (Mulrow, 1987). Obviously, if review articles do not routinely use well-founded criteria to identify, assess and synthesize information, review abstracts will be utterly deficient and will inaccurately and inefficiently convey information.

Another major finding is that there is also a lot of room for improvement at the paragraph level. Careful structure and sequence of ideas ought to be observed at this level, too. Paragraphs should be well-sequenced and have a clear link from one to the other. If a linguistically-marked paragraph structure is *not* strictly imposed by journal editors, authors should at least be urged to structure their abstracts in such a way that physical paragraphs correspond to conceptual ones. We would suggest that each paragraph correspond to one or two moves (for example, 1. purpose and methods; 2. results and 3. conclusions, and recommendations when they apply). Moreover, and bearing in mind non-native English speakers as well as 'non-expert' native speakers of English, connectors and/or special signalling devices could be recurred too, to explicitly indicate the hierarchical inter-relationships of the facts exposed. Indeed, it is well known that, when used appropriately, logical (that is, causal and relational) cohesion devices — which are almost non-existent in abstracts (Krvavac, 1987) — are valuable signposts to the readers, since they facilitate both their journey through the text and the integration of ideas in memory (Cooper, 1984: 133; Marshall, 1984: 94; Pearson and Camperell, 1985).

Earlier, we have also seen that some abstracts do not straightforwardly mention what 'was done or examined', but include an opening statement which sets the general field. This statement, which corresponds to the characteristic opening part of the typical article introduction (Swales, 1981; Hall et al., 1986) also serves the purpose of defining the authors' relationship towards the topic, indicating why they are interested in it or stating the shortcomings of previous work carried out in the field. It is interesting to note that, regardless of text- or research-type, this move is mostly encountered in general readership journals: it appeared 20 times in journals like *The British Journal of Medicine* or *The Lancet*, vs. 15 times in more specialized journals. This corroborates McKinley's finding (1983) that such a move appears more frequently in the introduction section of scholarly papers published in general readership journals than in that of articles appearing in specialized journals.

As already mentioned, guidelines for structured abstracts have been proposed for research papers with direct clinical implications and review articles only. Although Huth (1987a) holds that the possibilities for structured abstracts for case reports are less clear, we can suggest guidelines

which would highlight the vital elements of study design in this text-type. The format which seems most suitable would simply be: 1. case presentation, 2. findings and 3. comment. But, however simple this pattern is, our study indicates that it is far from being adopted as a general rule. In addition, we are of the opinion that structured abstracts should not be confined only to studies carried out in clinical research, but should be required in the other three research-types as well.

In the journals which do not require abstracts to be written with clearly labeled elements, the 'instructions for authors' section is too often very vague and nebulous. Some state that abstracts should be 'self-contained', 'informative, not factual', 'give factual essence of the article' or that they 'should be understood without access to the entire article'. Others simply mention that abstracts should limit themselves to a certain number of words (between 150 and 250). Along the same lines, instructions for abstract writing which can be found in style manuals for authors, editors and publishers in the biomedical sciences are not always very explicit either. Woodford (1983: 107), for example, recognizes that informative abstract writing is not an amateur affair but his 'advice' is no substantial help to the novice writer: 'an abstract is a non-critical, informative digest of the significant content and conclusions of the primary source material ... Informative abstracts clearly call for great skill on the part of the abstracter and are usually entrusted to professionals'. Such instructions do not emphasize the generation and coherent expression of ideas but rather remain at the level of rhetorical generality.

It could very well be argued that the structured format suggested by The Ad Hoc Working Group substantially lengthens conventional abstracts. True: an increase of around 48% has been mentioned (Mulrom et al., 1988), but qualitatively, the greater clarity of structured abstracts justifies the greater length. Each move is more detailed and better delineated and the necessary ones are present. It can thus be claimed that what is 'lost' on one side, (that is, space economy) is 'gained' on others because the proposed guidelines present potential advantages not only for readers but also for writers, searchers and reviewers. In the first place, many psychological studies have demonstrated the powerful effect of headings and sub-headings on comprehension (Dooling and Lachman, 1971; Bransford and McCarrell, 1974) and readability guidelines based on cognitive psychology advise using prominent headings (Huckin, 1982). Secondly, by providing writers with a framework that will help them to concisely organize and present the results of their investigation, the proposed format will force them to be more accurate, precise and ordered in their thinking and writing process and product. As for searchers of online bibliographic databases (MEDLAR or MEDLINE), the suggested guidelines will allow more

precise computerized literature searches and assist them to select relevant, valid and appropriate articles more quickly (Hewitt and Chalmers, 1985; L'Abbé et al., 1987). What is more, a scientist may be interested in one specific aspect of a research (for example, results, data synthesis). In an abstract structured with linguistically-marked paragraphs, s/he will not have to read the whole abstract — much less scan the entire article in search of the information s/he is looking for — but only the abstract section which is of interest to her/him. Lastly, as Huth (1987b) and Mulrow et al. (1988) note, the suggested guidelines will also facilitate peer review before publication. It would be interesting to carry out cross-linguistic studies on medical abstracts, similar to Fluck's research (1989) in German, to know whether the discursial pitfalls noted in Medical English abstracts are also observed in medical abstracts written in other languages and/or in other fields. If the answer to this question is positive, then the following questions arise: are the deficiencies similar to the ones observed in this study? If not, in what sense do they differ? Why? Here is groundwork for investigation in what is obviously a wide area for future research.

5. Conclusions

Our results must of course be interpreted in the light of the population we studied and, as was said before, should be further tested out. However, it can hardly be denied that the foregoing discussion showed that writing informative abstracts is no sloppy, amateur affair, and suggested that an important step is urgently needed to improve their scope, impact and quality. Writers and medical journal editors share their part of responsibility. On the one hand, the effectiveness with which scientists inform is crucial and this depends on the readability of their reports. It seems that medical scientists should be more sensitive to the whole process of negotiation of meaning, especially in abstract writing. They should do so with their intended readers' purpose in mind; otherwise they will fail to communicate effectively or even fail to communicate at all. On the other hand, as Huth (1987a) mentions, journal editors certainly take responsibility for the quality and importance of papers they select to publish; they have no less responsibility for the quality of abstracts. In the press of daily work, an editor cannot let inadequate abstracts slip into print. The 'instructions for authors' section should thus underscore the importance of a 'meticulously clear and systematically constructed abstract' (Mulrow et al., 1988: 615) so that the information contained in them be 'a clear and comprehensive reflection of the relevance and validity of the paper'. Ultimately, if a good fraction of today's residents in internal medicine

receive special training in the critical assessment of journal literature (Huth, 1989), it would be advisable to provide them with special training in informative abstract writing as well. As Namukwai and Williams (1988: 226) so adroitly point out: 'Vast amounts of time and money are spent in improving readers' reading ability. But if a fraction of this time and money were to be spent on improving writers' writing ability, the readers' task would be made so much easier and enjoyable.' Under those conditions, the answer to the question that Dr. Huth, Editor of *Annals of Internal Medicine*, raised in an article published two years ago: 'Can authors be helped to write abstracts that provide critically important evidence?' (1987b: 626) is definitely positive. With all this in mind, a further step towards full reading efficiency, effective, easier and enhanced readability of scientific knowledge and development could then be undertaken. The sooner the better.

Notes

- * This research was supported by grant N° M-310-88 from the Consejo de Desarrollo Científico, Humanístico y Tecnológico (CDCHT) de la Universidad de Los Andes (Mérida, Venezuela), and by the Consejo Nacional de Investigaciones Científicas y Tecnológicas (CONICIT).

Acknowledgments are due to Dr. Betty Lou Dubois and Dr. Abdel Fuenmayor for their constructive suggestions and comments on an earlier draft of this paper.

1. By 'operative research' is meant the 'how-to-do' kind of research (i.e., how to improve or solve 'technical' problems).
2. A preliminary study of 10 leading medical journals published last year indicated that research papers form 65% of the papers published, reviews about 20%, and case reports 15%.
3. Optional moves such as 'statement of the problem' or 'recommendations/suggestions' can of course be included.
4. In that respect, it is interesting to note that the Ad Hoc Working Group for the Appraisal of Medical Literature (1987), a group formed by 358 researchers from 118 different countries, suggested that key aspects of purpose, methods and results be consistently stated *in this order* and in a standardized manner with explicitly-marked headings (8 for research papers and 6 for review articles), with a partially controlled vocabulary and in a point rather than in a narrative form.

References

A. Medical Literature

- Ad Hoc Working Group for Critical Appraisal of The Medical Literature (1987). A proposal for more informative abstracts of clinical articles *Annals of Internal Medicine*. 106: 598-604.
- Altman, D.G. and Gardner, J.M. (1987). More informative abstracts. *Annals of Internal Medicine*. 107: 790-791.

- Berry, E.M. (1981). The evolution of scientific and medical journals. *New England Journal of Medicine*. 305 (7): 400–402.
- Brewer, J.I. (1987). Summary of analysis of published papers. *American Journal of Obstetrics and Gynecology*. 156 (3): 526–526.
- Currie, B.F. (1975). The importance of medical periodicals for the Continuing education of physicians. *Association of Hospital Medical Education Journal*. 7: 1–2, N° 3.
- (1976). Continuing education from medical periodicals. *Journal of Medical Education*. 51: 420.
- Harris, J.J. (1966). Survey of medical communication sources available for continuing physician education. *Journal of Medical Education*. 41: 737–755.
- Hewitt, P., and Chalmers, T.C. (1985). Using MEDLINE to peruse the literature. *Controlled Clinical Trials*. 103: 802–816.
- Huth, E.J. (1987a). Structured abstracts for papers reporting clinical trials. *Annals of Internal Medicine*. 106: (4): 626–627.
- (1987b). Keeping up with the medical literature: reprint available. *Annals of Internal Medicine*. 106: (4): 627.
- (1989). The underused medical literature. *Annals of Internal Medicine*. 110 (2): 99–100.
- International Committee of Medical Journal Editors. (1988). Uniform requirements for manuscripts submitted to biomedical journals. *Annals of Internal Medicine*. 108 (2): 258–265.
- L'Abbé, K.A., Detsky, A.S., and O'Rourke, K. (1987). Meta-analysis in clinical research. *Annals of Internal Medicine*. 107: 224–233.
- Light, R.J., and Pillemer, D.B. (1984). *Summing up: the Science of Reviewing Research*. Cambridge, MA: Harvard University Press.
- Lock, S. (1988). Structured abstracts. *British Medical Journal*. 297: 156.
- Louis Harris and Associates (1987). *The Future of Information Systems for the Medical Sciences: A study Conducted for The New York Academy of Medicine*. New York: Louis Harris and Associates.
- Maher, J. (1987). English as an international language of medicine. *Medical Education*. 21: 283–284.
- Mulrow, C.D. (1987). The medical review article: State of the science. *Annals of Internal Medicine*. 106: 485–488.
- , Thacker, S.B., and Pugh, J.A. (1988). A proposal for more informative abstracts of review articles. *Annals of Internal Medicine*. 108: 613–615.
- Politser, P.E. (1987). Medical education for a changing future: new concepts for revising texts. *Medical Education* 21: 320–333.
- Vollan, D.D. (1955). *Postgraduate medical education in the United States*. Chicago: American Medical Association.
- Williamson, J.W., German, P.S., Weiss, R., Skinner, E.A., and Bowes, F. (1989). Health science information management and continuing education of physicians: a survey of US primary care practitioners and their opinion leaders. *Annals of Internal Medicine*. 110: 151–160.
- Woodford, F.P. (ed.) (1983). *Scientific Writing for Graduate Students. A Manual on the Teaching of Scientific Writing*. Bethesda, MO: Council of Biology Editors.

B. Linguistic and Psycholinguistic Literature

- Bransford, J.D., and McCarrell, H.S. (1974). A sketch of a cognitive approach to comprehension. In *Cognition and the Symbolic Processes*. W.B. Weimer, and D.S. Palermo, (eds.), 189–229. Hillsdale, NJ: Lawrence Erlbaum.

- Carrell, P.L. (1985). Facilitating ESL reading comprehension in teaching text structure. *TESOL Quarterly*. 19: 727-752.
- (1987). Readability in ESL. *Reading in a Foreign Language*. 4 (1): 21-40.
- Cooper, M. (1984). Linguistic competence of practised and unpracticed non-native readers of English. In *Reading in a Foreign Language*. C. Alderson, and A. H. Uquhart (eds.), 123-132. London: Longman.
- Cremmins, E.T. (1982). *The Art of Abstracting*. Philadelphia: ISI Press.
- Davies, F. (1985). Toward a classroom-based methodology for identifying information structures in text. In *Reading for Professional Purposes: Methods and Materials in Teaching Languages*. J. Ulijn, and A.K. Pugh (eds.), 103-120. Leuven: Acco.
- Dooling, D.J., and Lachman, R. (1971). Effects of comprehension on retention of prose. *Journal of Experimental Psychology*. 88: 216-222.
- Dudley-Evans, T. (1986). *Genre Analysis: an Investigation of the Instruction and Discussion Sections of MSc. Dissertations*. LSU. ESP Collection. N° 3.5H/16.
- Ertl, N. (1969). A new way of documenting scientific data from medical publications. *Karger Gazette*. 27 (20): 1-3.
- Fluck, H.R. (1989). Uergleichende Analyse deutschsprachiger Abstracts in wissenschaftlichen Zeitschriften. In *Special Language: From Human Thinking to Thinking Machines*. C. Laurén and M. Nordman (eds.), 291-307. Clevedon: Multilingual Matters.
- Gopnik, M. (1972). *Linguistic Structures in Scientific Texts*. The Hague: Mouton.
- Graetz, N. (1985). Teaching EFL students to extract structural information from abstracts. In *Reading for Professional Purposes: Methods and Materials in Teaching Languages*. J.M. Ulijn, and A.K. Pugh (eds.), 123-135. Leuven: Acco.
- Guagfu, Z. (n.d.). Tense, voice and sentence length and complexity in abstracts. Xian Jiaotong University. Unpublished manuscript.
- Hall, D., Hawkey, R., Kenny, B., and Storer, G. (1986). Patterns of thought in scientific writing: a course in information structuring for engineering students. *English for Specific Purposes*. 5 (2): 147-160.
- Hancock, J. (1989). Learning with data bases. *Journal of Reading*. 32 (7): 582-586.
- Hartley, A. (1988). Computer-stimulated acquisition of reading ability in LSP. In *Beads or Bracelet? How Do We Approach LSP*. A.M. Cornu, J. Vanparijs, M. Delahaye, and L. Baten, (eds.), 348-356. Leuven: Oxford University Press.
- Harvey, A. (1986). A macrostructure analysis of 20-computer originated abstracts in the field of engineering. In *Seminar on Research into Languages for Specific Purposes: A Report*. G. Latorre, (ed.), 25-34. University of Santiago, Chile. Jan. 6-7.
- , and Horsella, M. (1988). Exploring and exploiting the structure of computational abstracts. Paper presented at the IVth Latin American ESP Colloquium. Sao Paolo, Brazil.
- Huckin, T. (1982). A cognitive approach to readability. In *New Essays in Technical and Scientific Communication: Theory, Research and Criticism*. P. Anderson, and J. Brockman, (eds.), Farmingdale, NY: Baywood Press.
- Huerta, T., Ibañez, I. and Kaulen, A. (1986). Balancing institutional and motivational factors in ESP syllabus design. *English for Specific Purposes*. 5 (2): 189-195.
- Kenny, B. (1985). Pattern of writing or pattern of thought? Paper presented at the International Conference on ESP, Colombo.
- Kieras, D.E., and Bovair, S. (1981). *Strategies for abstracting main idea from simple technical prose* (Technical Report N° 10). University of Arizona.
- Kintsch, W., and Keenan, J.M. (1973). Reading rate and retention as a function of the number of propositions in the base structure of sentences. *Cognitive Psychology*. 5: 81-90.

- , and van Dijk, T.A. (1983). *Strategies in Discourse Comprehension*. New York: Academic Press.
- , and Yarbrough, J.C. (1982). Role of rhetorical structure in text comprehension. *Journal of Educational Psychology*. Vol. 74, p. 828–834.
- Krvavac, A. (1987). Procedure in abstract writing. Paper presented at the VIth European Symposium on LSP. Vaasa: Finland. August.
- Mandler, J.M. (1978). A code in the node: the use of a story schema in retrieval. *Discourse Processes*. 1: 14–35.
- Marland, M. (1981). *Information skills in the secondary curriculum*. London: Methuen Educational.
- Marshall, N. (1984). Discourse analysis as a guide for informal assessment of comprehension. In *Promoting Reading Comprehension*. J. Flood, (ed.), 79–97. Newark, Delaware: IRA.
- McKinlay, J. (1983). *An Analysis of the Discussion Section of Medical Journal Articles*. LSU/ESP Collection (3.4/B) University of Birmingham.
- Meyer, B.J.F. (1977a). The structure of prose: effects on learning and memory and implications for educational practice. In *Schooling and the Acquisition of Knowledge*. R.C. Anderson, R.J., Spiro, and W.E. Montague, (eds.), 179–200. Hillsdale, NJ: Lawrence Erlbaum Associates.
- (1977b). What is remembered from prose: a function of passage structure. In *Discourse Production and Comprehension*. R.O. Freedle, (ed.), 41–58. Norwood, NJ: Ablex Publishing Corporation.
- (1980). Signaling in text and its interactions with reader strategies. Paper presented at a meeting of the American Educational Research Association. Boston, April.
- (1981). Basic research on prose comprehension: a critical review. In *Comprehension and the Competent Reader: Interspeciality Perspective*. D.F. Fisher, and C.W. Peters, (eds.). New York: Praeger.
- , Brandt, D.M. and Bluth, G.J. (1980). Use of top-level structure in text: key for reading comprehension of ninth-grade students. *Reading Research Quarterly*. 16: 72–103.
- Namukwai, V., and Williams, R. (1988). The readability of government reports: a case study from Zambia. *Reading in a Foreign Language*. 5 (1): 205–229.
- Pearson, P.D., and Camperell, K. (1985). Comprehension of text structures. In *Theoretical Models of Processes of Reading*. Singer, H. and R.B. Ruddell, (eds.), 323–342. 3rd ed. Newark, Delaware: International Reading Association.
- Swales, J. (1981). *Aspects of article introductions*. Aston ESP Research Report N° 1. The University of Aston. Birmingham.
- Ulijn, J. (1985). A present state of LSP reading research as reflected by a recent symposium. In *Reading for Professional Purposes: Methods and Materials for Teaching Languages*. J. Ulijn, and A.K. Pugh, (eds.), 12–25. Leuven: Acco.
- van Dijk, T.A. (1980). *Macrostructures*. Hillsdale, NJ: Erlbaum.
- , and Kintsch, W. (1983). *Strategies of Discourse Comprehension*. New York: Academic Press.

Françoise Salager-Meyer holds a B.A. and a M.A. in Russian linguistics (University of Lyons, France), and a Ph.D. in Foreign Language Education from the University of Texas at Austin (USA). She taught French and Russian in the United States and is currently teaching English and Russian at the University of The Andes, Mérida (Venezuela). She has published several articles on the linguistic, rhetorical and discourse analysis of medical prose in *System*, *Reading in a Foreign Language*, *Forum*, *EMP Newsletter*, *The ESP Journal*, *Interface*, *Estudios de Lingüística Aplicada*.